Problems In Group Theory

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Contents

Chapter 1

Basic Group Theory

1.1 Introduction

Problem 1.1.1. *Determine which of the following binary operations are associative:*

1. *the operation on $\mathbb{Z}$ defined by $a * b = a - b$*
2. *the operation on $\mathbb{R}$ defined by $a * b = a + b + ab$*
3. *the operation on $\mathbb{Q}$ defined by $a * b = \frac{a+b}{5}$*
4. *the operation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) * (c, d) = (ad + bc, bd)$*
5. *the operation on $\mathbb{Q} - \{0\}$ defined by $a * b = \frac{a}{b}$*

Proof. This exercise is pretty trivial as we just have to compute $a * (b * c) = (a * b) * c$. For 1., $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$ but $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$. Rest of them are pretty similar. $\square$

Problem 1.1.2. *Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative.*

Proof. By definition of residue addition,

$$[a] + [b] = [a + b]$$

Therefore, we get the following.

$$\begin{aligned} [a] + ([b] + [c]) &= [a] + [b + c] \\ &= [a + (b + c)] \\ &= [(a + b) + c] \\ &= [a + b] + [c] \\ &= ([a] + [b]) + [c] \end{aligned}$$

$\square$

Problem 1.1.3. *Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative.*

Proof. By definition of residue multiplication,

$$[a] \cdot [b] = [a \cdot b]$$

Therefore, we get the following.

$$\begin{aligned} [a] \cdot ([b] \cdot [c]) &= [a] \cdot [b \cdot c] \\ &= [a \cdot (b \cdot c)] \\ &= [(a \cdot b) \cdot c] \\ &= [a \cdot b] \cdot [c] \\ &= ([a] \cdot [b]) \cdot [c] \end{aligned}$$

□

Problem 1.1.4. *Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.*

Proof. There is no multiplicative inverse for $[a]$ with $\gcd(a, n) > 1$. The proof for it pretty trivial, if there was a inverse, it would mean that

$$a \cdot b \equiv 1 \pmod{n} \implies 0 \equiv 1 \pmod{\gcd(a, n)}$$

That's a contradiction. □